



E-waste & it's management

A Seminar

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in partial fulfilment for the award of the degree

of

Bachelor of Technology

in

Electrical & Electronics Engineering

At



CT INSTITUTE OF TECHNOLOGY, JALANDHAR

December 2020

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Introduction. .

Electronics waste, commonly known as e-scrap or e waste, is the trash we generate from surplus, broken, and obsolete electronic devices. Electronics contains various toxic and hazardous chemicals and materials that are released into the environment if we do not dispose of them properly. E-waste or electronics recycling is the process of recovering material from old devices to use in new products.



In 2010, the United States *disposed of some*

2,439,000 TONS

of waste generated from *electronic devices*.

2,500,000

2,000,000

1,500,000

1,000,000

500,000

0

649,000

of these tons were recycled,

while an astounding

1,790,000

were thrown in landfills or
incinerated.^[1]

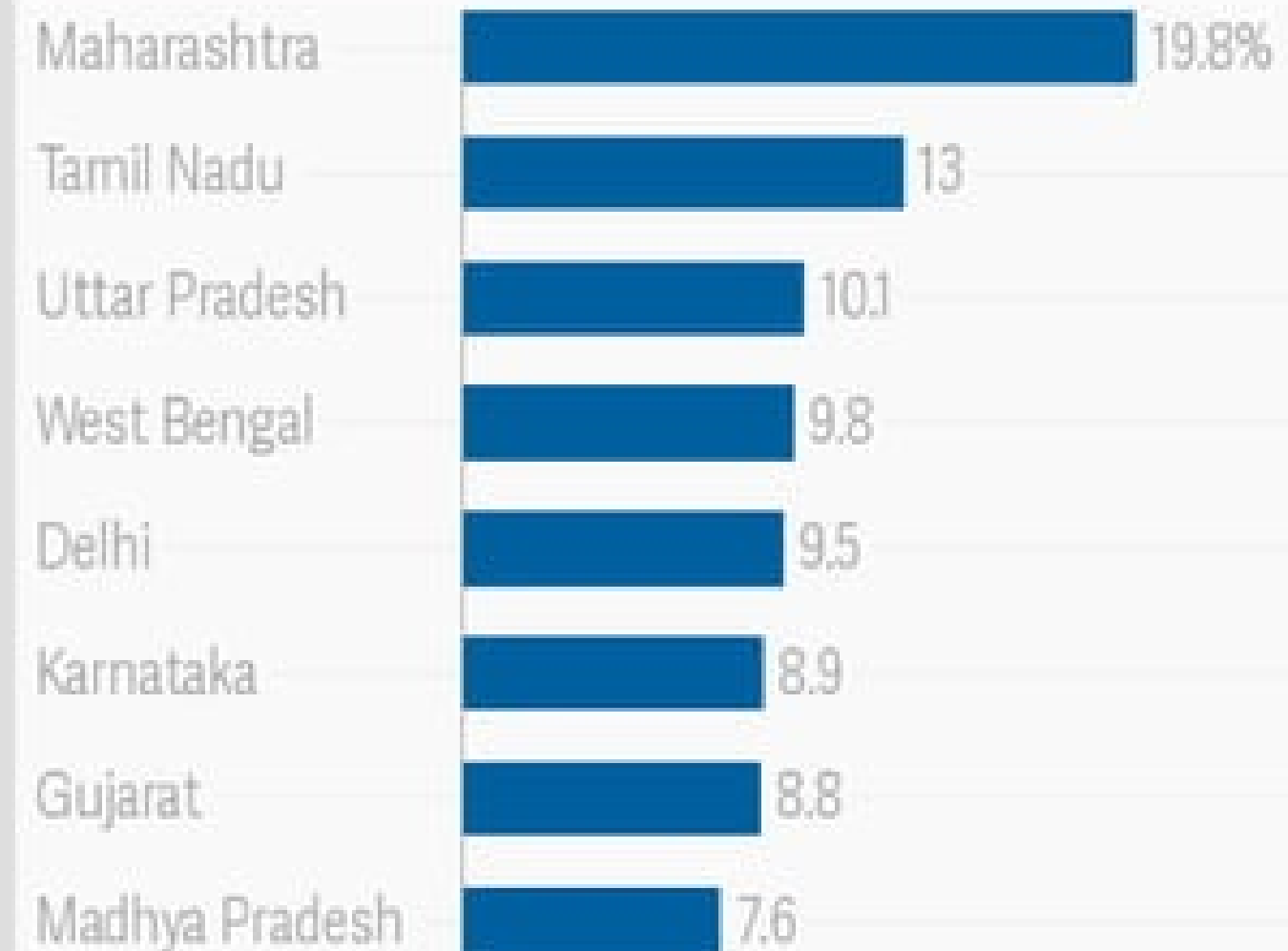


Biggest contribution of E-waste in India. .

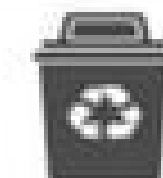
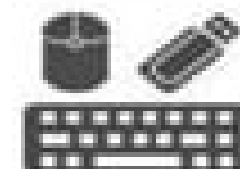
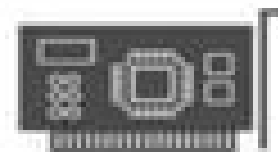
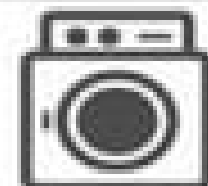
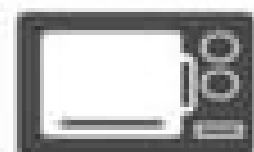
Over 2 Million old PCs ready for Disposal in India



State wise contribution of E-waste in India



This graph is from QUARTZ INDIA article by Ananya Bhattacharya



E-waste fact. .

There are More Mobile
Phones Than People. .

People Discard 350,000
Mobile Phones Daily. .

DO YOU
KNOW



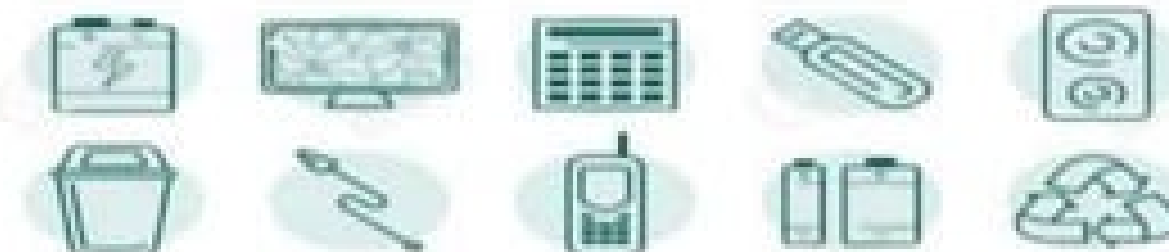
Current Scenario in India. .

The total E-waste in India has been estimated to be **1,86,180 Tonnes per year.**



E-waste Piling Up

City Name	E-waste (in Tonnes)
Mumbai	11,017
Delhi	9,730
Bangalore	4648
Chennai	4132
Kolkata	4025
Ahmadabad	3287
Hyderabad	2833
Pune	2584
Surat	1836



Growth of E-waste in India. .



: Growth of Ewaste in India



Quantity of WEEE (Waste Electrical and Electronic Equipment) generated in Indian States. .

SN	States	WEEE (Tones)	SN	States	WEEE (Tonnes)
1	Andaman and Nicobar Islands	92.2	19	Lakshadweep	7.4
2	Andra Pradesh	12,780.3	20	Madhya Pradesh	7,800.6
3	Arunachal Pradesh	131.7	21	Maharashtra	20,270.6
4	Assam	2,176.7	22	Manipur	231.7
5	Bihar	3,055.6	23	Meghalaya	211.6
6	Chandigarh	359.7	24	Mizoram	79.6
7	Chhattisgarh	2,149.9	25	Nagaland	145.1
8	Dadra and Nagar Haveli	29.4	26	Orissa	2,937.8
9	Daman and Diu	40.8	27	Puducherry	284.2
10	Delhi	9,729.2	28	Punjab	6,958.5
11	Goa	427.4	29	Rajasthan	6,326.9
12	Gujarat	8,994.3	30	Sikkim	78.1
13	Haryana	4,506.9	31	Tamil Nadu	13,486.2
14	Himachal Pradesh	1,595.1	32	Tripura	378.3
15	Jammu and Kashmir	1,521.5	33	Uttar Pradesh	10,381.1
16	Jharkhand	2,021.6	34	Uttarakhand	1,641.1
17	Karnataka	9,118.7	35	West Bengal	10,059.4
18	Kerala	6,171.8		Total	146,180.7

□ Country Level WEEE assessment study by IRGSSA (2005)

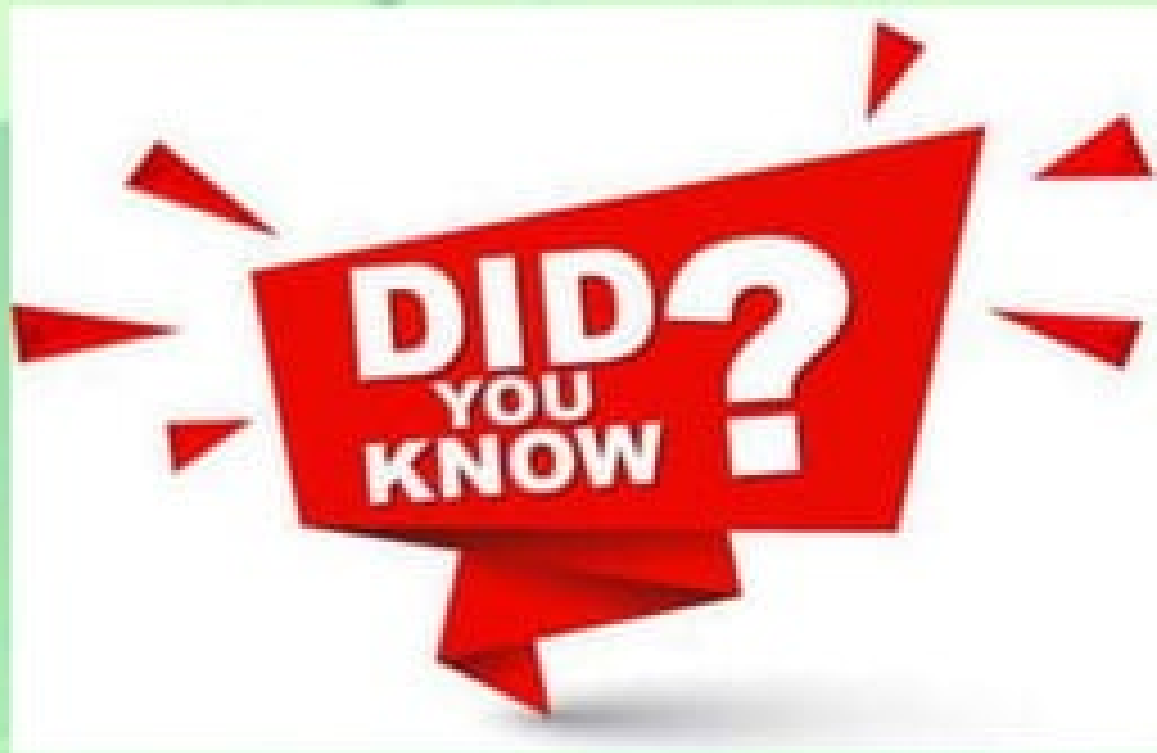
TOP 5 E-WASTE GENERATING COUNTRIES IN THE WORLD

USA	CHINA	JAPAN	GERMANY	INDIA
11.7 Million Tonnes	6.17 Million Tonnes	2.2 Million Tonnes	2.0 Million Tonnes	1.8 Million Tonnes

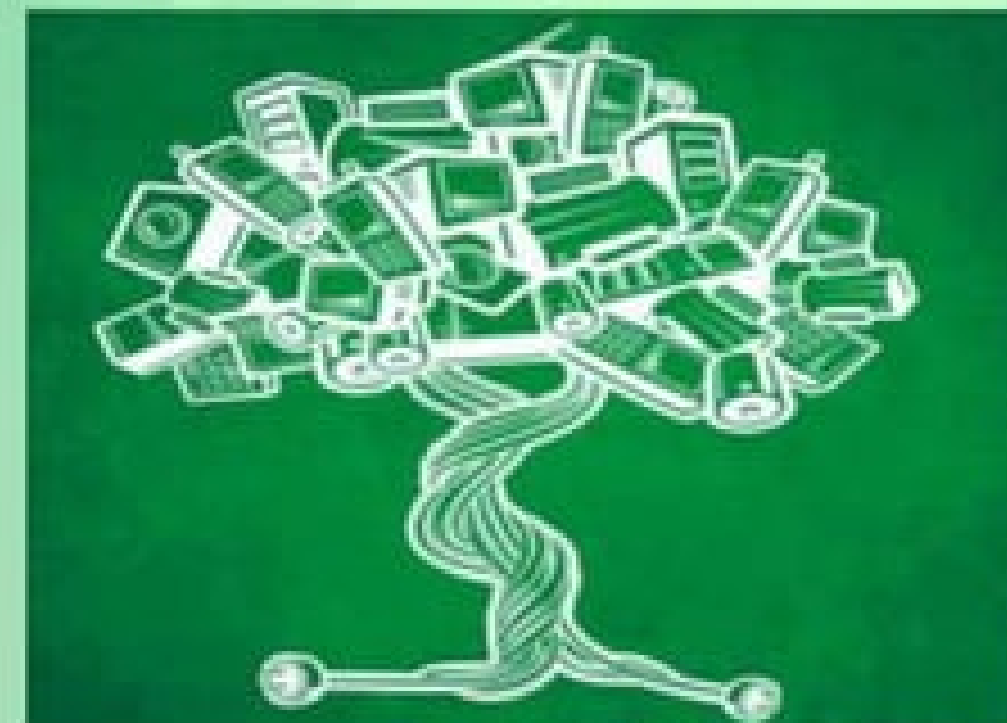
❑ India was ranked as the 5th largest generator of electronic waste in the world. A study conducted by The Associated Chambers of Commerce and Industry of India and KPMG in 2016, ranked India among one of the top 5 countries in the world in terms of e-waste generation, with an estimated 1.85 million tonnes generated annually.

❑ According to a recent United Nations University (UNU) report, global e-waste will increase by 38% in the decade between 2020 and 2030.

E-waste fact. .



The largest E-waste site on earth is on **GUIYU, China**.
Approximately **88% of Children** there
have Dangerous levels of lead in there
blood.

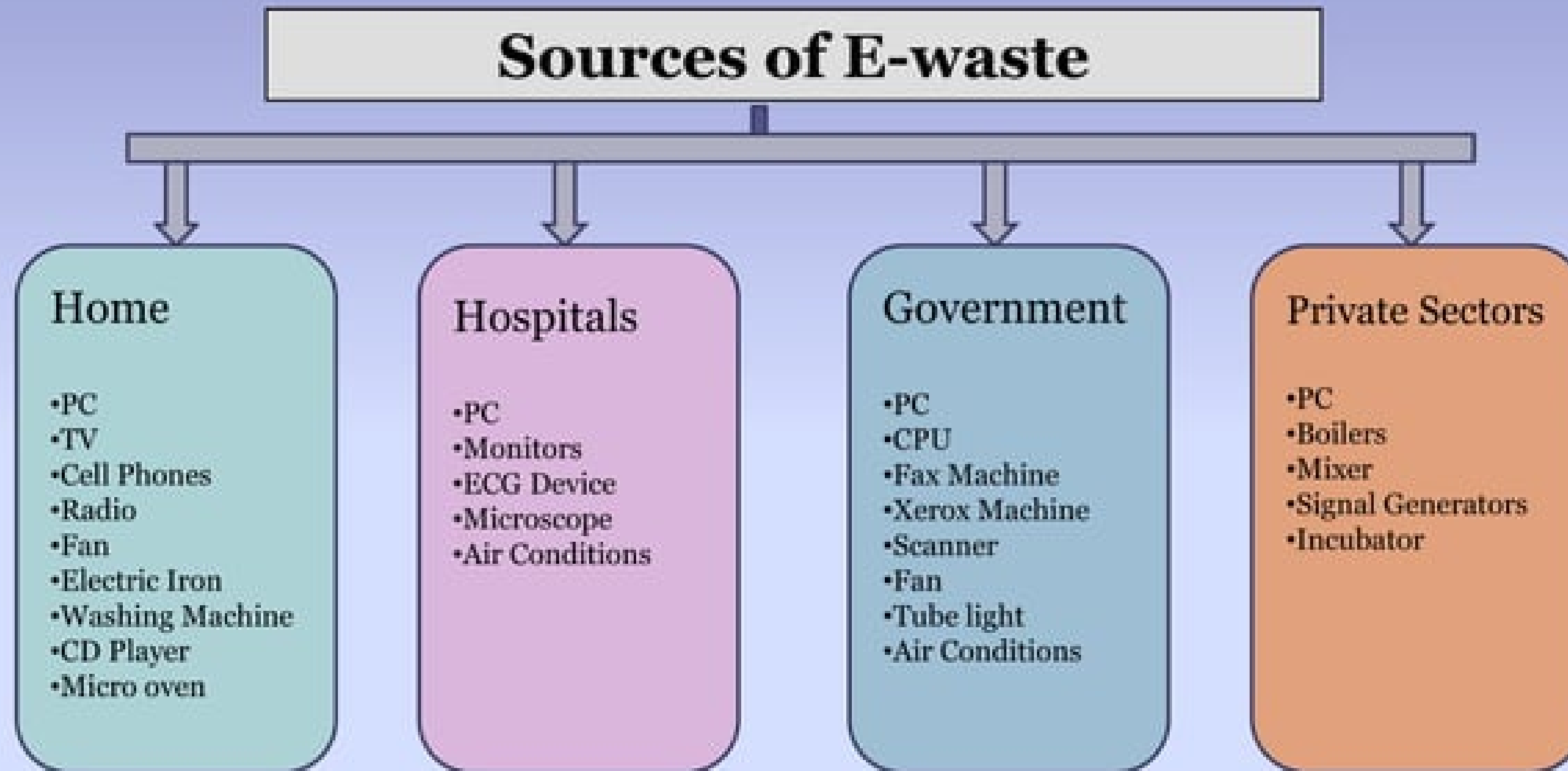


E-waste. .

- Electronic waste or E-waste is any broken or unwanted electrical or electronic appliance.
- E-waste includes computers, consumer electronics, phones, medical equipments, toys and other items that have been discarded by their original users.
- E-waste also include waste which is generated during manufacturing or assembling of such equipments.

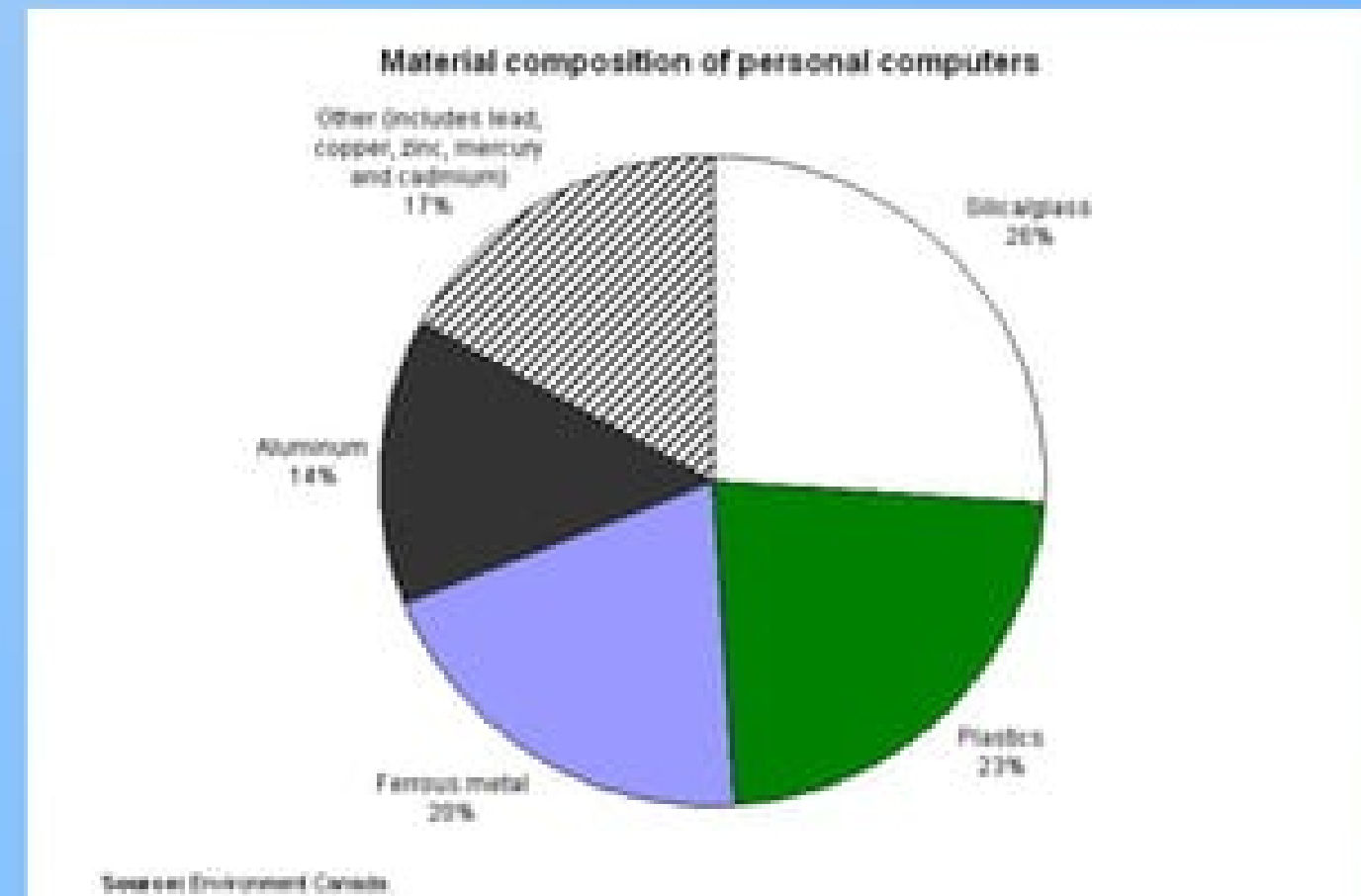


Sources of E-waste..



Challenges. .

- E-waste contains several different substances and chemicals, many of which are toxic and are likely to create adverse impact on environment and health, if not handled properly.
- E-waste is the fastest growing waste stream.
- Technological advances speed up obsolescence & lead to more e-waste.



Impact. .

Effects on Human Health

- Affects brain development of children.
- Chronic damage to the brain.
- Respiratory and skin disorders due to bioaccumulation in fishes.
- Asthmatic bronchitis.
- Reproductive and developmental problems.
- Lung Cancer.
- Damage to heart, liver and spleen.
- Damage to central and peripheral nervous systems, blood systems and kidney damage.

Effects on Environment

- Pollution of Ground-Water.
- Acidification of soil.
- Air Pollution.
- E-Waste accounts for 40 percent of the lead and 75 percent of the heavy metals found in landfills.



Element of E-waste effect on human body..

Element	Effect on human body.
Lead	Damage to central and peripheral nervous system, blood and kidney damage. Affect brain development of children.
Chromium	Asthmatic Bronchitis. DNA damage.
Cadmium	Toxic irreversible effect on human health. Accumulates in kidney and liver. Causes neural damage.
Mercury	Chronic damage to brain and respiratory system.
Plastics including PVC	Burning produces dioxin. It causes reproductive and developmental problems ; Immune system damage.

EWASTE



E-waste fact. .

Computers, mobile phones and other electronic products use a staggering 320 tones of gold and more than 7,500 tones of silver annually world wide.

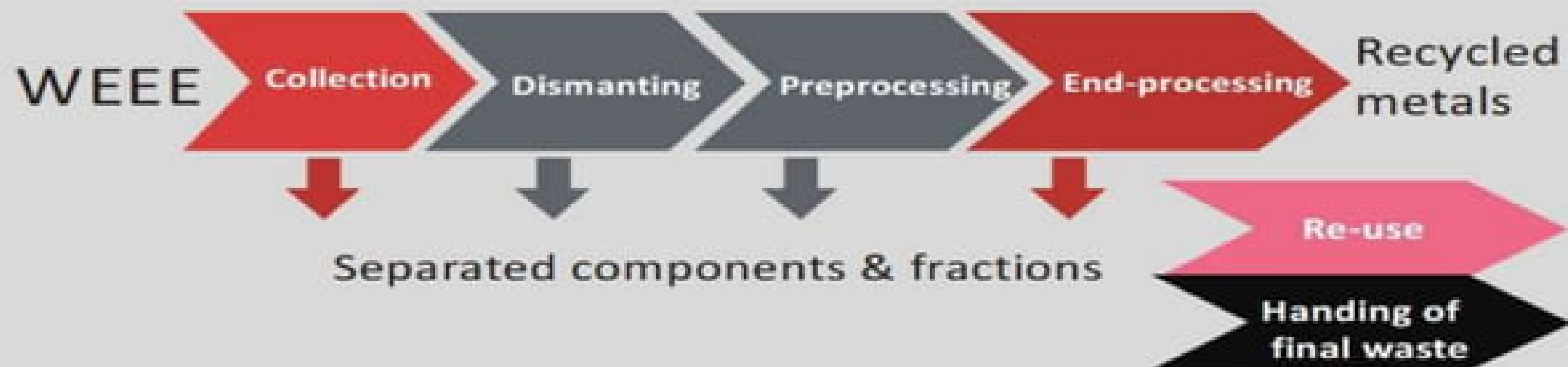
Did YOU
Know?



E-waste Management. .

E-waste management should begin at the point of generation. This can be done by:-

- Waste minimization techniques by sustainable product design.
- Regulatory Compliance and Follow Agreed Standards.



Source: Recycling – From E-waste to Resources: Sustainable Innovation and Technology Transfer Industrial Sector Studies, UNEP, 2009

Steps for E-waste management. .

- Step 1: Use 3R technique to minimize the E-waste.
- Step 2 Identify the E-waste category item.
- Step 3: Identify the E-waste composition or determine it.
- Step 4: Identify possible hazardous content in E- waste.
- Step 5: Identify, whether the E-waste component is hazardous or the entire E-waste item is hazardous.



E-waste management. .



To use things with care to reduce the amount of waste generated.



To repeat use of items or parts of items.



To use waste as resources.

Recycling of E-waste. .

- **Disassembly/dismantling :-**

Disassembly is the systematic removal of components, parts, a group of parts or a subassembly from a product which is in E-waste.

- **Upgrading :-**

It includes comminuting and separation of materials using mechanical / physical and/or metallurgical processing. Methods to recover materials include incineration and refining.

- **Materials Recovery :-**

The material are recovered by recycling facilities. The plastic, glass, metals can be recovered by sorting them before mixing with other waste.

Advantages of recycling W-waste. .

Asset recovery

Reduction of need for landfill

Reduction of junks and clutters

Resale and Reuse

Creation of jobs

**We can be
a Super
Hero ??**



Conclusion. .

Reduce

Recover



Reuse

Recycle

= Zero Land Fill

Implement Proper E-waste Disposal. .

One man's 'junk' is other man's
'Treasure' !

(Anon)



References. .

❑ **Web Articles/Journals:-**

- E-waste scenario in India, its management and implications By:- Sushant B. Wath, P. S. Dutt & T. Chakrabarti – 18 Jan 2010
- E-waste management in India : A study of current scenario By:- Dr. Neha Garg & Deepak Kumar Adhana – January 2019
- E-waste in India - Research Unit (LARRDIS) Rajya Sabha Secretariat, New Delhi, By:- Dr. V. K. Agnihotri, Secretary- General, Rajya Sabha – June, 2011
- Electronic Waste and India By:- Dr. S. Chatterjee

❑ **Web-sites:-**

- <https://www.ewaste1.com>
- https://en.wikipedia.org/wiki/Electronic_waste
- <https://www.drishtiias.com/printpdf/increasing-e-waste>

❑ **Links:-**

- <https://blogs.furman.edu/mgoldsmith2/background-of-e-waste/>
- <https://www.dtnext.in/News/TopNews/2018/11/21015819/1096445/Tamil-Nadu-grapples-with-ewaste-challenge.vpf>
- https://www.slideshare.net/vbvklone/e-waste-27139823?qid=e1ab9b00-99be-4516-9a95-3e363bab479c&v=&b=&from_search=25

Today's Leisure
Will Become Tomorrow's Junk



THANK YOU!!!!